1. A water pipe system comprising three pipes connects two reservoirs, A and B, and a pond C (Figure 1). The flow directions in pipes 1 and 3 are as indicated. The head losses in pipes 1 and 3 are the same, $h_1 = h_3$. Neglect the head loss at the pipe junction. The density of water $\rho = 1000$ kg/m^3, and the gravitational constant $g = 9.81$ m/s^2. Determine:

(a) The head loss in pipe 1, h_1. (20 marks)

(b) The flow direction and head loss in pipe 2. (20 marks)

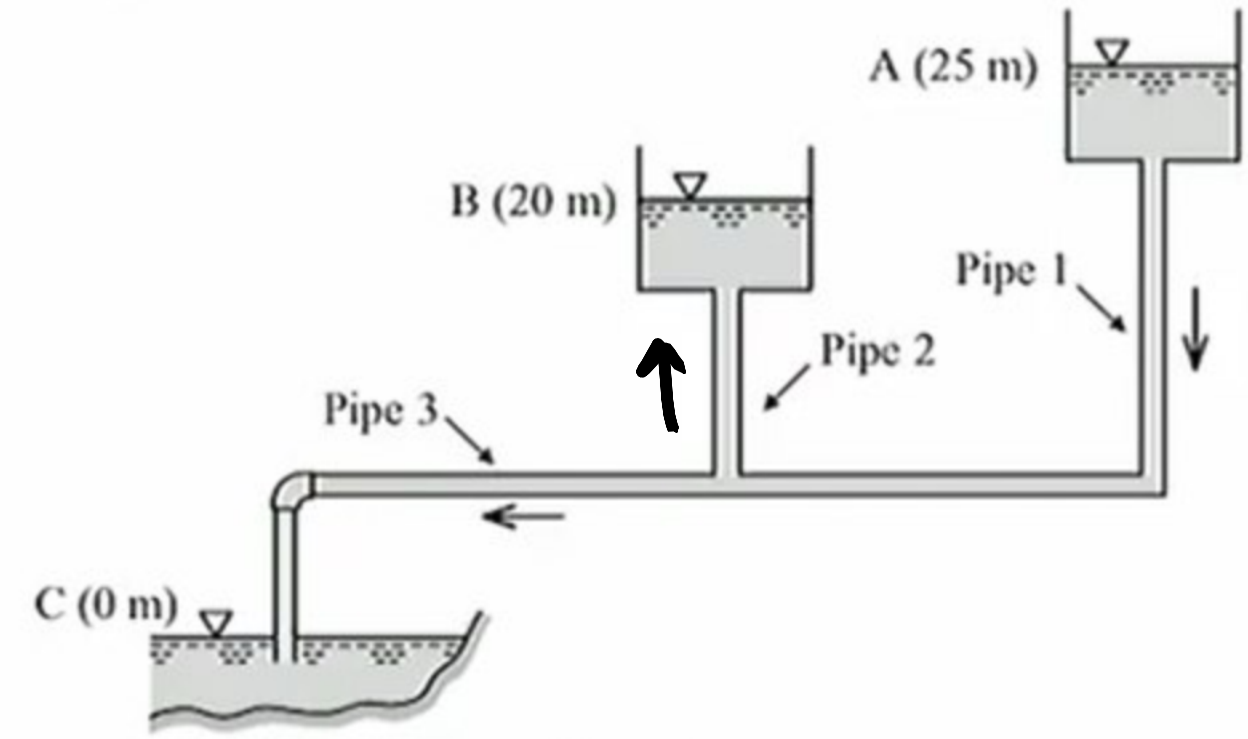


Figure 1

a) $$\cancel{\frac{P_A}{\rho g}} + \cancel{\frac{V_A^2}{2g}} + z_1 = \cancel{\frac{P_C}{\rho g}} + \cancel{\frac{V_C^2}{2g}} + \cancel{z_3} + h_1 + h_3$$

$$25 = 2h_1 \quad \because h_1 = h_3$$

$$h_1 = 12.5\,\text{m}$$

b) $$\cancel{\frac{P_A}{\rho g}} + \cancel{\frac{V_A^2}{2g}} + z_1 = \cancel{\frac{P_B}{\rho g}} + \cancel{\frac{V_B^2}{2g}} + z_2 + h_1 + h_2$$

$$25 = 20 + 12.5 + h_2$$

$$h_2 = -7.5\,\text{m}$$

$\therefore$ The flow diaection of pipe 2 is upwards.

2. An air jet flows out from a rectangular opening located on the surface of a control volume (CV) (Figure 2). The opening has a length L and a width W (into the plane of the paper). In the inertial frame, the air velocity varies linearly along the y-direction only. At Points A and B, the air velocities are V and 0, respectively. The air density is denoted by ρ.

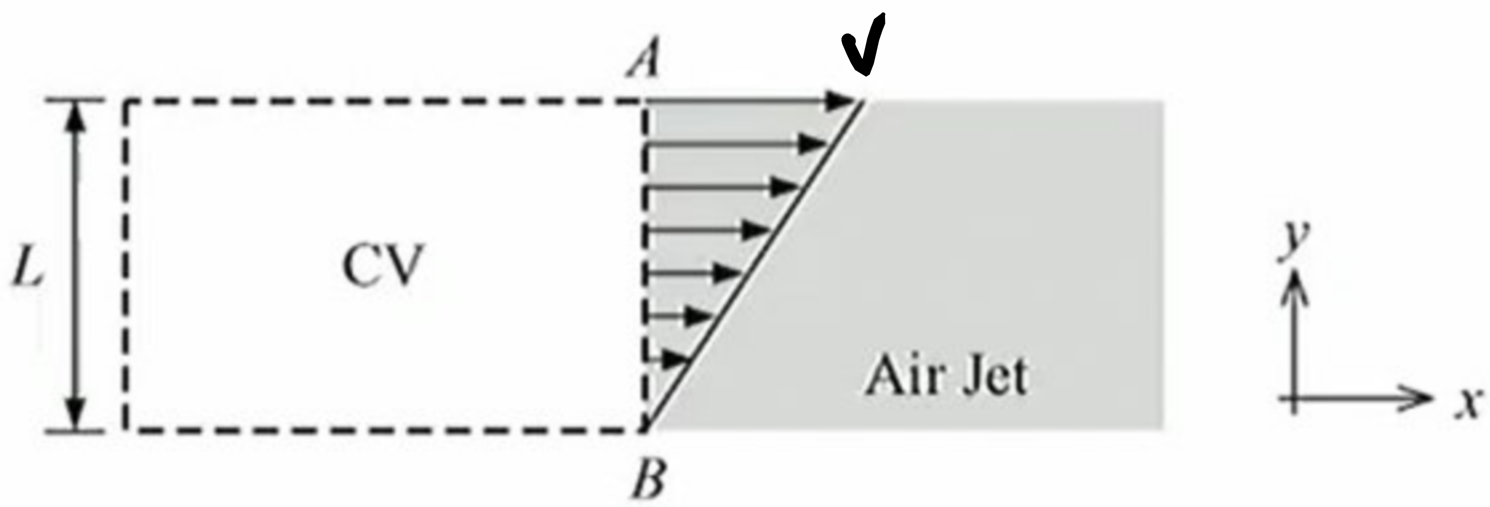


Figure 2

(a) If the CV is stationary, fill in the two blanks using the given variables:

The average flow velocity of air at the CV outlet is $\frac{V}{2}$.

(5 marks)

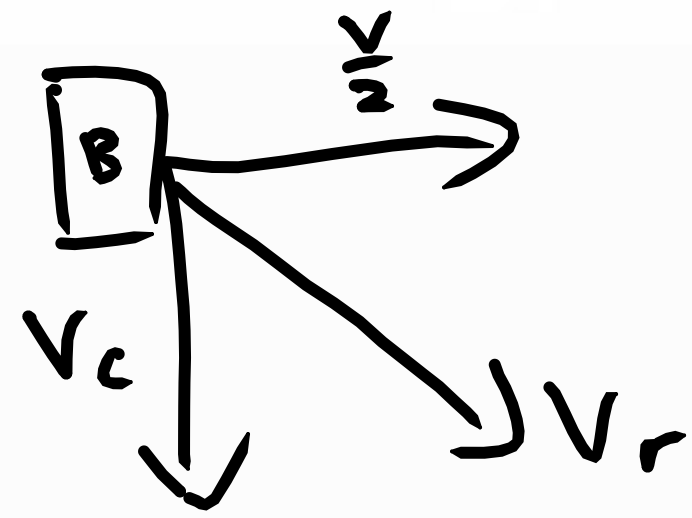
Hence, the mass flow rate of air at the CV outlet is $\frac{\rho L W V}{2}$.

(5 marks)

$$\dot{m} = \rho A V$$
$$= \rho L W \left(\frac{V}{2}\right)$$
$$= \frac{\rho L W V}{2}$$

(b) If the CV is moving in the positive y-direction at a constant speed V_c in the inertia frame and the air speed remains unchanged in the inertia frame, fill in the two blanks using the given variables:

In the frame relative to the CV, what is the direction of the y-component of the air velocity at the CV outlet?



$\therefore V_{r_y} = -V_c$ y-direction

The mass flow rate of air at the CV outlet is $\frac{\rho L W V}{2}$ $\because \frac{V}{2} \perp CV$